Rishabh Vaibhav

rishabhvaibhav7@gmail.com +91 7061864929 linkedin.com/in/rishabh-vaibhav github.com/rishabhvaibhav

Professional Summary

I am an Immediate Joiner with 4 years of experience building production-grade ETL pipelines, LLM-integrated data workflows, scalable data platforms, and intelligent automation systems for clients including Google, AuditBoard, Bank of America, and Airbus. Specialized in RAG pipeline architecture, vector database integration, LLM-based entity extraction, Gen AI-powered data transformation, multi-agent systems, and large-scale web data acquisition. Processed 1TB+ monthly data with 98–99.2% schema compliance. Architected 200+ automated data ingestion systems and 12+ scalable ETL pipelines. Built POCs for clients leveraging Azure OpenAI, AWS Bedrock, and LangChain. Currently at Zocket building 15+ platform scrapers and an LLM-powered ETL on Apache Airflow, ClickHouse, and Neo4j. Proficient in Python, PySpark, SQL, Django, FastAPI, and cloud platforms (Azure, AWS, GCP). Azure Certified (AZ-204). Seeking Data Engineer, Python Developer, Gen AI Developer, Data Scientist, and Data Acquisition Engineer roles.

Experience

Zocket Technologies *SDE II* Sep 2026 – Nov 2026

Product: BrandOS | Chennai, India

Project: BrandOS – Brand Intelligence Data Platform

- Built multi-platform brand-monitoring scrapers for 15+ platforms (Amazon, Flipkart, Myntra, Meesho, Google Maps, Play Store, App Store, X/Twitter, Instagram, LinkedIn, Reddit, YouTube) into one unified schema using Playwright, httpx, and lxml
- Replaced paid Apify and Firecrawl actors with in-house X/Twitter and Instagram scrapers capturing posts, nested comments, and engagement metrics; cut third-party scraping cost
- Handled Akamai and Cloudflare anti-bot protection with real Chrome profiles, TLS fingerprint impersonation (curl_cffi), session management, and Bright Data proxy failover; built an in-house proxy service classifying residential vs. datacenter IPs
- Built incremental Amazon and Flipkart product and review crawlers using JSON-LD rating snapshots and newest-first pagination; reruns cost one page load per unchanged product
- Built BrandOS ETL ingesting ∼20 sources per brand, classifying data points with LLMs (OpenAI, Gemini) into a 16-layer taxonomy, and loading signals into ClickHouse and a Neo4j knowledge graph
- Deployed Apache Airflow 3.2 on AWS EKS (Helm, KubernetesExecutor, git-sync); wrote DAGs for S3 Parquet → ClickHouse loads and social-media ETL in a bronze/silver/gold medallion model
- Shipped a FastAPI crawling service with Docker and GitHub Actions CI/CD to AWS (ECR, S3, IAM); added pytest suites, canary checks, and Sentry monitoring

AzimuthGRC *Data Engineer II* Mar 2026 – Aug 2026

Domain: GRC & Compliance | Gurugram, India

Project: Global Statutes & Regulations Data Pipeline

- Built end-to-end ETL pipeline to ingest Statutes and Regulations from legal sources across multiple global jurisdictions; automated extraction, transformation, and loading into Azure SQL Database with Pydantic schema validation
- Designed Python data models to represent regulatory document structure and relationships; ensured schema consistency and integrity across diverse source formats for downstream LLM consumption
- Deployed data to production with Delta versioning to track document changes and revisions over time; maintained full audit trail for regulatory updates and compliance reporting
- Performed large-scale PDF extraction across 3,000+ pages using PyMuPDF and Unstructured; parsed documents in hierarchical order preserving section, subsection, and clause structure for RAG-based retrieval

GlobalLogic *Data Engineer* Oct 2024 – Mar 2026

Client: Google | Hyderabad, India

Project: Government Schemes Data Pipeline (LLM-Powered)

- Built LLM-powered data ingestion pipeline extracting government scheme records across dynamic websites using Selenium, Playwright, and BeautifulSoup with anti-bot handling and proxy rotation
- Transformed unstructured HTML into structured JSON; built 12+ scalable ETL pipelines using Python, Pydantic, and Google Cloud Platform (GCP) services, improving processing speed by 40% through schema validation and deduplication
- Integrated LangChain with Duckie (internal LLM) for automated entity extraction and data enrichment; lowered error rate from 8% to 0.5% and achieved 99.2% data accuracy
- Stored 1M+ enriched records in Knowledge Graph with optimized indexing and semantic relationships; improved downstream query performance by 60% for ML model consumption
- Fine-tuned and aligned Google AI model to real-world source data using LoRA; improved result quality and factual accuracy across government scheme entities for search and discovery
- Published enriched records to Knowledge Panel (KP) via semantic network; enabled structured entity representation and LLM-

based retrieval for internal search systems

- Monitored pipeline health via SC Studio with CI/CD workflows and automated alerting; minimized unplanned downtime by 90%

Client: AuditBoard | Gurugram, India

Project: Regulatory Data Scraper Engine (RAG + Gen AI Powered)

- Architected RAG-ready data pipeline ingesting 5M+ compliance documents from 200+ regulatory websites (UK, US, global) using BFS/DFS crawling with Requests, BeautifulSoup, and Playwright
- Deployed scraping infrastructure on Azure VMs with EC2 instances running cron jobs for scheduled extraction; handled anti-bot bypass, CAPTCHA solving, and rotating proxies
- Designed end-to-end ETL pipeline covering extraction, cleaning, Pydantic validation, and loading into Azure; maintained 98% schema compliance across 2TB+ monthly data
- Integrated Azure OpenAI and LangChain for LLM-based classification, entity extraction, and semantic tagging; reduced false positives by 65% and improved document targeting precision
- Built Python data models for regulatory document hierarchy; implemented Delta versioning to track document changes and enable temporal queries for compliance audits
- Created Flet-based Python desktop UI for real-time scraper monitoring and control; enabled 3-person team to manage 400+ scrapers with minimal oversight
- Orchestrated 200+ website workflows via n8n with retry logic and deduplication; eliminated 320+ hours/month of manual data engineering work
- Managed end-to-end source code and release workflows on GitLab; maintained branching strategy and CI/CD version control

Project: Search CAC Insights – Centralized Metadata Platform

- Centralized metadata repository tracking 25+ BD teams, 100+ organizations, and $50M+ spend
- Administered 150+ process names, pipeline metadata, and feed locations; enabled collaboration for 50+ stakeholders across data engineering and analytics teams
- Analyzed 200+ data feeds and generated automated summaries; supported executive decisions and shortened reporting cycle

ALTEN India *Data Engineer* Jul 2022 – Sep 2024

Client: Airbus | Bangalore, India

Project: Flight Delay Root-Cause Analytics Platform

- Designed and built an end-to-end ETL data pipeline in Python, integrating flight operations data and weather data via REST APIs to enable root-cause analysis of flight delays across multiple airlines and aircraft models for Airbus
- Processed 50GB+ of historical flight and weather data using PySpark for the initial bulk load, then transitioned to incremental weekly data ingestion, reducing processing time and compute resource usage for ongoing updates
- Engineered geospatial data processing logic using latitude/longitude coordinates to reconstruct flight paths and perform point-by-point weather correlation analysis, identifying storm, fog, and wind impacts along each route
- Performed data profiling, cleaning, transformation, and feature engineering using PySpark, SQL, Pandas, and NumPy to classify delays as weather-related (departure/en-route/arrival) vs. mechanical/operational
- Optimized large-scale data processing using PySpark DataFrame API with partitioning and efficient join strategies to handle high-volume flight and weather record joins
- Implemented caching and memory optimization techniques in the Django backend, significantly reducing dashboard load times and improving REST API response speed for 200+ concurrent users
- Built a Django-based interactive analytics dashboard with geospatial map visualizations – flight routes, delay hotspots, and weather overlays – enabling 200+ Airbus engineers and analysts to identify delay patterns instantly
- Enabled cross-airline, cross-aircraft-model delay pattern analysis by aggregating otherwise siloed airline data, supporting data-driven recommendations for aircraft design, maintenance scheduling, and operational efficiency

Key Projects

RAG Pipeline for Regulatory Compliance Search

- Built end-to-end RAG system using LangChain + Azure OpenAI + FAISS vector embeddings; enabled semantic search across 5M+ compliance documents with 94% retrieval accuracy
- Implemented chunking strategies, embedding generation, and similarity search with Euclidean distance and cosine similarity for legal document retrieval; reduced manual legal research time by 70%
- Integrated hallucination control mechanisms and prompt engineering best practices to ensure factual accuracy in generated responses

LLM-Based Automated Entity Extraction Platform

- Designed extraction pipeline using Pydantic + LangChain + Azure OpenAI for unstructured document processing; reduced false positives by 65% and automated 320+ hours/month of manual data entry

- Integrated with Knowledge Graph for semantic entity storage; improved downstream ML model query performance by 60%
- Built Streamlit-based POC for client demos showcasing real-time entity extraction and classification capabilities

Multi-Agent Chatbot System with AWS Bedrock Integration

- Architected a multi-agent chatbot system using LangChain and AWS Bedrock for intelligent document Q&A and automated data processing workflows
- Implemented semantic similarity search with FAISS and pgvector for context-aware retrieval; optimized embedding models for domain-specific accuracy
- Developed prompt engineering framework with hallucination control, chain-of-thought reasoning, and output validation for reliable AI responses
- Delivered POC to client demonstrating automated report generation and intelligent data summarization capabilities

Mobile Application Data Extraction Framework

- Built automated data extraction framework for mobile applications using Appium, Selenium, and AJAX-based dynamic content scraping
- Implemented caching layer with Redis and request deduplication to optimize extraction speed and reduce redundant API calls
- Integrated Firecrawl and Crawl4AI for intelligent JavaScript-rendered content extraction from modern web and mobile interfaces
- Developed logging and monitoring pipeline with structured error tracking for real-time debugging and performance optimization

Technical Skills

LLM, Gen AI & AI Agents: LangChain, LlamaIndex, Azure OpenAI, OpenAI API, Google Gemini, AWS Bedrock, LLM Fine-tuning (LoRA, QLoRA), RAG Pipelines, Prompt Engineering, Hallucination Control, Chain-of-Thought Reasoning, Multi-Agent Systems, Chatbot Development, Embeddings, Hugging Face Transformers, LLM-based Entity Extraction, LLM Classification, Semantic Similarity, Cosine Similarity, Euclidean Distance

Vector Databases & Semantic Search: Pinecone, ChromaDB, FAISS, pgvector, Weaviate, Neo4j, Knowledge Graph, Semantic Search, Vector Embeddings, Similarity Metrics

Data Engineering: ETL/ELT Pipelines, PySpark, Pandas, NumPy, PyArrow, Parquet, Medallion Architecture (Bronze/Silver/Gold), Delta Versioning, Schema Validation, Pydantic, Data Modeling, Apache Spark, Feature Engineering, Data Profiling, Caching, Memory Optimization, Incremental Data Ingestion, Geospatial Data Processing

Orchestration, MLOps & DevOps: Apache Airflow 3 (KubernetesExecutor), n8n, CI/CD, Docker, Docker Compose, Helm, MLflow, GitHub Actions, GitLab CI/CD, pytest, Sentry, Model Monitoring, Workflow Automation, Logging, Structured Error Tracking

Cloud & Infrastructure: Azure (Databricks, OpenAI, SQL Database, Data Factory, VM), AWS (S3, EKS, ECR, IAM, RDS, EC2, SageMaker, Bedrock), GCP, Kubernetes (EKS), Linux VPS (Hetzner), MinIO

Web Scraping, Data Acquisition & Mobile Extraction: Selenium, Playwright, Requests, httpx, curl_cffi, BeautifulSoup, lxml, XPath, Firecrawl, Apify, Crawl4AI, Scrapy, BFS/DFS Crawling, Incremental Crawling, Anti-bot Handling (Akamai, Cloudflare), Browser & TLS Fingerprinting, Rotating Proxies, Bright Data, CAPTCHA Handling, Retry Logic, Deduplication, AJAX Scraping, Appium, Mobile Application Extraction, Dynamic Content Rendering

Databases, Caching & Data Warehouses: ClickHouse, Neo4j, PostgreSQL, MySQL, SQL Server, Azure SQL Database, AWS RDS, Snowflake, BigQuery, MongoDB, Redis, pgvector

Languages, Frameworks & APIs: Python (asyncio), SQL, Cypher, PySpark, Django, FastAPI, Pydantic, REST APIs, Node.js, Shell Scripting, Streamlit, Flet, AJAX, Geospatial Libraries

PDF Processing & Document Parsing: PyMuPDF, Unstructured, Hierarchical Document Parsing, OCR, Large-Scale Document Extraction

Monitoring, Visualization & UI Development: Streamlit, Flet, Django Dashboard, SC Studio, Grafana, REST APIs, n8n, Geospatial Map Visualizations, Interactive Dashboards, POC Development, Client Demos

Education

Haldia Institute of Technology — *B.Tech, Computer Science & Engineering* — 2018 – 2022

CGPA: 7.9/10

Certifications

Microsoft Certified: Azure Developer Associate (AZ-204)